

Yuming Yan, PH.D.

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EMPLOYMENT

- 04/2025 - **Postdoctoral Researcher**
Elmore Family School of Electrical and Computer Engineering
Purdue University, West Lafayette, IN, USA
Web: [Electric machine](#) | [Power electronics](#) | [Wearable robots](#)
- 04/2024 - 04/2025 **Postdoctoral Research Fellow**
Department of Mechanical and Aerospace Engineering
Joint UNC/NCSU Biomedical Engineering Department
North Carolina State University, Raleigh, NC, USA

EDUCATION

- 07/2019 – 03/2024 **Ph.D. in Electrical Engineering**
Nanyang Technological University, Singapore
- 08/2013 – 06/2017 **B.E. in Electrical Engineering**
Xi'an Jiaotong University, Xi'an, China

RESEARCH GRANT EXPERIENCE

- 2026 **Department of Energy (DOE), AI-Enabled Demagnetization and Safe Removal of Permanent Magnets from Electric Motor Rotors for Recycling**
Contribution: Theoretical modeling, mathematical derivations, and technical figure preparation.
- 2024 **National Science Foundation (NSF), CPS: Efficient Human-in-the-Loop Learning for Human-Centric Cyber-Physical Systems**
Contribution: authored the methodology section and contributed to proposal writing
Total award: \$350,000 over three years
- 2023 **National Institutes of Health (NIH), Smart and Connected Health: Spine-Hip Exoskeletons With Learning-Based Optimal Control for Low Back Pain Alleviation**
Contribution: literature review and methodology development
Award: \$274,000 per year for four years
- 2023 **Panasonic Corporation, Lightweight PCB Motor With High Specific Power**
Contribution: methodology section, literature review, and proposal development
- 2022 **Myron Zucker Grant Committee, IEEE Industry Applications Society, Integrated Winding Concept for Variable Flux Memory Machine**
Contribution: methodology section and proposal development

RESEARCH INTERESTS

I develop application-driven electric machines and integrated actuation systems by connecting electromagnetic design with mechanical structures, controls, and system-level validation. My research spans sustainable and ultra-efficient electric machines, high-specific-torque direct-drive actuators, and human-centered robotic systems for energy-efficient infrastructure, electrified mobility, and human mobility.

PEER REVIEWED PUBLICATIONS

9 journal publications and 7 conference publications including 4 first-author journals and 3 first-author conferences.

Journals (9)

- J1. **Y. Yan**, J. S. Huang, J. Zhu, Z. Hou, W. Gao, I. Lopez-Sanchez, N. Srinivasan, A. Srihari, and H. Su, "Compact and Foldable Hip Exoskeleton With High Torque Density Actuator for Walking and Stair-Climbing Assistance in Young and Older Adults," *IEEE/ASME Transactions on Mechatronics*, vol. 31, no. 4, pp. 4467-4477, Aug. 2026. (impact factor 6.3)
- J2. **Y. Yan**, S. Cai, J. Chen, S. Wang, B. Cheong, C. J. Gajanayake, A. K. Gupta, and C. H. T. Lee, "Design and Analysis of Memory Machine With Asymmetric Delta-Array Consequent Pole for Energy-Efficient Vehicular Applications," *IEEE Transactions on Energy Conversion*, vol. 40, no. 3, pp. 1696-1711, Sep. 2025. (impact factor 6.1)
- J3. **Y. Yan**, F. Shen, S. Wang, B. Cheong, C. J. Gajanayake, A. K. Gupta, S. Cai, X. Yuan, J. Chen, S. Xie, and C. H. T. Lee, "Novel Memory Motor Drive With Integrated Winding Concept for Traction Applications," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 6, pp. 7171-7183, Jun. 2023. (impact factor 7.5)
- J4. **Y. Yan**, Y. Fan, F. Shen, H. Chen, S. Cai, X. Yuan, and C. H. T. Lee, "Hybrid-Magnet Variable Flux Memory Machine With Improved Field Regulation Capability for Electric Vehicle Applications," *IEEE Transactions on Transportation Electrification*, vol. 9, no. 1, pp. 586-597, Mar. 2023. (impact factor 8.5)

- J5. H. Kim, **Y. Yan**, J. Im, M. Fereydoonian, K. Lee, A. Dornala, A. Halawa, J. S. Lee, and W. Lee, "Functional Integration of Power Electronics and Electric Machines Enabling High-Power-Density Electric Drives," IEEE Transactions on Transportation Electrification, Early Access, 2026. (impact factor 8.5)
- J6. F. Shen, **Y. Yan**, S. Wang, B. Cheong, C. J. Gajanayake, A. Gupta, and C. H. T. Lee, "Investigation of a Flux-Switching Machine With U-V-Array Permanent-Magnet Arrangement for Traction Applications," IEEE Transactions on Vehicular Technology, vol. 72, no. 10, pp. 12580–12591, Oct. 2023. (impact factor 7.5)
- J7. S. Xie, S. Cai, Y. Zuo, L. Cao, J. Zhu, A. Li, **Y. Yan**, and C. H. T. Lee, "A New Hybrid Concentrated-Winding Concept With Improved Power Factor for Permanent-Magnet Vernier Machine," IEEE Transactions on Industrial Electronics, vol. 70, no. 11, pp. 11109–11120, Nov. 2023. (impact factor 7.4)
- J8. Y. He, G. Yang, S. Xie, F. Shen, **Y. Yan**, and C. H. T. Lee, "Magnetic Equivalent Circuit Model for a Two-Degree-of-Freedom Rotary-Linear Machine With Transverse-Flux Structure," IEEE Transactions on Transportation Electrification, vol. 10, no. 4, pp. 10338–10348, Dec. 2024. (impact factor 8.5)
- J9. H. Li, S. Zhao, M. Xia, S. He, Q. Yang, **Y. Yan**, and H. Zhao, "Spontaneous Formation of Non-Uniform Double Helices for Elastic Rods Under Torsion," Physics Letters A, vol. 381, p. 689, 2017. (impact factor 2.7)

Peer Reviewed Conference Proceeding Papers (7)

- C1. **Y. Yan**, S. D. Sudhoff, T. Chin, and W. Lee, "Integrated Motor-Pump with a Spanwise-Varying Multi-Airfoil Rotor for Cooling of AI Datacenters and Edge Compute Nodes", IEEE Energy Conversion Congress and Exposition (ECCE), Vancouver, Canada, 2026. Accepted.
- C2. **Y. Yan**, H. Kim, and M. W. Lee, "Analysis and Design of Line-Start Permanent-Magnet-Assisted Synchronous Reluctance Machine for Aircraft Electric Hydraulic System," IEEE Transportation Electrification Conference & Expo (ITEC), Novi, MI, USA. Invited oral presentation.
- C3. **Y. Yan**, F. Shen, S. Ramakrishna, G. C. Jayampathi, S. Wang, and C. H. T. Lee, "A Novel Fault-Tolerant Flux-Switching Memory Machine With High Flux Controllability," IECON 2021-47th Annual Conference of the IEEE Industrial Electronics Society, Toronto, Canada, pp. 1-6, 2021.
- C4. F. Shen, Y. Zuo, **Y. Yan**, and C. H. T. Lee, "Comparative Study of Dual-Permanent-Magnet Machines With Different Consequent-Pole Stators," IEEE Energy Conversion Congress and Exposition (ECCE), Nashville, USA, pp. 4596-4601, 2023.
- C5. F. Shen, **Y. Yan**, S. Ramakrishna, C. J. Gajanayake, S. Wang, and C. H. T. Lee, "Comparative Study of Flux-Switching Machines With T-Array and U-Array Permanent-Magnet Arrangements," IEEE Energy Conversion Congress and Exposition (ECCE), Detroit, USA, pp. 1-6, 2022.
- C6. F. Shen, **Y. Yan**, B. C. S. Onn, C. J. Gajanayake, S. Wang, and C. H. T. Lee, "Analysis of a Vernier Machine With Spoke-V Array Permanent Magnets," IECON 2022-48th Annual Conference of the IEEE Industrial Electronics Society, Brussels, Belgium, pp. 1-6, 2022.
- C7. F. Shen, **Y. Yan**, S. Ramakrishna, C. J. Gajanayake, S. Wang, and C. H. T. Lee, "Analysis of Vernier Machine With Stator-V-Shaped Permanent-Magnet Arrangement," IECON 2021-47th Annual Conference of the IEEE Industrial Electronics Society, Toronto, Canada, pp. 1-6, 2021.

Conference Abstracts and Workshops Papers (2)

- A1. J. Zhou, W. Gao, **Y. Yan**, J. S. Huang, Z. Hou, I. Lopez-Sanche, and H. Su, "Experiment-Free Reinforcement Learning for an Unobtrusive and Compact Hip Exoskeleton with High-Torque-Density Actuators," Northeast Bioengineering Conference (NEBEC), Brooklyn, NY, 2025.
- A2. J. Zhou, W. Zhou, W. Gao, **Y. Yan**, Z. Xu, and H. Su, "A High Torque Density and Highly Compliant 7-DOF Collaborative Robotic Arm," IEEE/ASME (AIM) International Conference on Advanced Intelligent Mechatronics, Boston, MA, 2024.

MANUSCRIPTS UNDER REVIEW

- M1. **Y. Yan**, S. Cai, X. Yuan, J. Chen, and C. H. T. Lee, "Improved Variable-Flux Permanent-Magnet Machine With DC-Biased Hybrid Excitation for Variable-Speed Safety-Critical Applications," IEEE Transactions on Vehicular Technology (under 1st round major revision).
- M2. **Y. Yan**, H. Kim, and W. Lee, "Analysis and Design of an IE6-Class Line-Start Permanent-Magnet-Assisted Synchronous Reluctance Machine," IEEE Transactions on Energy Conversion (under review).
- M3. **Y. Yan**, J. Im, and W. Lee, "Toward a Closed-Loop Supply Chain for NdFeB Magnets in Electric Vehicles: A Review of End-of-Life Recovery Technologies and Process Metrics", IEEE Applied Power Electronics Conference and Exposition (APEC), 2027, (under review).
- M4. **Y. Yan**, and W. Lee, "Multi-Physics Design of Integrated Electrical Pump for AI Data Center Cooling," IEEE/ASME Transactions on Mechatronics (in preparation).
- M5. **Y. Yan** and H. Su, "Fully Direct-Drive Hip Exoskeleton with High Backdrivability and High Bandwidth," IEEE Transactions on Robotics (in preparation).
- M6. **Y. Yan** and W. Lee, "Design Optimization and Comparative Study of a Line-Start Electric Machine for Aircraft Electro-Hydraulic System," IEEE Transactions on Industrial Electronics (in preparation).

TEACHING ADVISING AND MENTORING

Co-Instructor and Project Mentor, Purdue University

- 2026 Fall **Invited lecturer, ECE 32100 - Electromechanical Motion Devices, Purdue University**
Classroom lecturing, laboratory instruction, and hands-on student guidance.
- 2023 Fall **Co-Instructor and Project Mentor, EE6503 Modern Electrical Drives, Nanyang Technological University**
Presented lectures and mentored 10 student teams in electric machine design.
Organized presentations and technical reports; evaluated and provided feedback on student work.

Teaching Assistant

- 2024 **Teaching Assistant, MAE 435 Principles of Automatic Control, North Carolina State University**
Undergraduate research mentoring in wearable exoskeletons, classroom integration of lab-developed hip exoskeletons and hands-on student training.
- 2023 **Teaching Assistant, EE2005 Electrical Devices and Machines, Nanyang Technological University**

	Organized student presentations and evaluated assignments.
2022	Teaching Assistant, EE4207 Control System Design, Nanyang Technological University Facilitated discussion-based tutorials for groups of 15 students; organized presentations and evaluated assignments.
2022	Teaching Assistant, EE4504 Design of Clean Energy Systems, Nanyang Technological University Facilitated tutorials and supervised experimental sessions for groups of 20 students; organized presentations and evaluated assignments.
2022	Teaching Assistant, EE4534 Modern Distribution Systems With Renewable Resources, Nanyang Technological University Supervised team-based experiments for groups of 20 students; organized presentations and evaluated experimental results.

Senior Design (7)

- Jin Sen Huang, Hip exoskeleton with high-torque-density actuation for walking and stair-climbing assistance in young and older adults, Ph.D. student, North Carolina State University, Apr. 2024-Apr. 2025.
- Chengming Liu, Fault-tolerant electric motor drive for aerospace applications, Nanyang Technological University, Fall 2023.
- Vinith Kumar S/O Vijaya Kumaran, High-performance fault-tolerant electric motor for aerospace applications, Nanyang Technological University, Spring 2022.
- Yuteng Yan, DC-biased current control for motor systems, M.S. student, Nanyang Technological University, Spring 2022.
- Kenzhi Iskandar Wong, Hybrid-excited motor for aerospace applications, Nanyang Technological University, Spring 2022.
- Weiyang Cheng, High-performance linear generator for wave-energy conversion, M.S. student, Nanyang Technological University, Fall 2022.
- Muyu Chen, Fault-tolerant electric motors for small electric vehicles, M.S. student, Nanyang Technological University, Fall 2021.

Mentored Undergraduate Research (3)

- Giridhar Balasubramaniam, Integrated electrical pump design for AI data center cooling, undergraduate research assistant, Purdue University, Fall 2025.
- Juncheng Zhou, High-torque-density, highly compliant 7-DOF collaborative robotic arm, M.S. student at Rice University and research assistant at North Carolina State University, Fall 2024.
- William Zhou, Unmanned ground vehicle system with direct-drive electric motors, high school student, North Carolina State University, Fall 2024.

INVITED TALKS AND CONFERENCE PRESENTATIONS

2026	IEEE Energy Conversion Congress and Exposition (ECCE), Vancouver, Canada.
2026	IEEE Transportation Electrification Conference & Expo (ITEC), Novi, MI, USA.
2023	Rolls-Royce RRE-Singapore Knowledge Series, Rolls-Royce internal
2022	Rolls-Royce Global Committee on Machine Electromagnetics Meeting, Rolls-Royce internal
2021	IEEE Industrial Electronics Society Annual Conference (IECON), Toronto, Canada.

AWARDS

2026 Fall	Conference Travel Funds, Purdue University
2019-2024	Fully Funded Research Scholarship, Nanyang Technological University
2022	Finalist, Three-Minute Thesis Presentation Symposium, Nanyang Technological University
2016	Siyuan Scholarship, Xi'an Jiaotong University
2015	Siyuan Scholarship, Xi'an Jiaotong University
2014	Outstanding Student Award, Xi'an Jiaotong University

PROFESSIONAL SERVICE

Technical Reviews / Publications Referee

- IEEE/ASME Transactions on Mechatronics
- IEEE Transactions on Energy Conversion
- IEEE Transactions on Transportation Electrification
- IEEE Transactions on Vehicular Technology
- IEEE Transactions on Industrial Electronics
- IEEE Transactions on Power Electronics
- IEEE Journal of Emerging and Selected Topics in Power Electronics

Conference Reviewer and Organizer Work

- International Conference on Electrical Machines and Systems (ICEMS), 2023
- IEEE Industrial Electronics Society Annual Conference (IECON), 2023
- International Council on Electrical Engineering Conference (ICEE), 2023
- Organizing Committee, Three-Minute Thesis Presentation Symposium, Nanyang Technological University, 2021

Professional Society Memberships

- Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, American Society of Mechanical Engineers (ASME)
- IEEE Robotics and Automation Society
- IEEE Industry Applications Society
- IEEE Industrial Electronics Society

REFERENCES

Prof. Matt Woongkul Lee, Assistant Professor

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Prof. Hao Su, Associate Professor

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